Proposition 7.1 (Double sums of powers decomposition)**.** *For non-negative integers n, m, and an arbitrary integer t,*

$$\Sigma^2\, n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[+\binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2}\Sigma^0\, n^0 - \binom{1-t+\frac{k}{2}}{k+1}\Sigma^1\, n^0$$
$$+\binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2}\Sigma^0\, n^0 - \binom{0-t+\frac{k}{2}}{k+1}\Sigma^1\, n^0\Bigg]\delta^k t^m.$$

For example, for $\Sigma^2\, n^2$ we have